Other-Directed Emotion Regulation and Sensitivity to Fairness

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Loading data

Loading in the .csv files with participant demographic, behavioral (e.g., EROS scores), and neuroimaging data (e.g., extracted cluster values). If you are interested in the data, please email.

```
# Load .csvs with subject data
full_df <- read_csv("/Users/tur61139/Documents/GitHub/rf1-kos/df_MRdata_n138.csv")
```

```
## Rows: 12494 Columns: 81
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (7): race, ethnicity, sub_gender, gender_at_birth, endowment_label, co...
## dbl  (73): sub_id, tppid, sub_age, nTrial, run, cue_Onset, event_duration, b...
## date  (1): sub_dob
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df <- read_csv("/Users/tur61139/Documents/GitHub/rf1-kos/df_MRcovar_n138.csv")
```

```
## Rows: 138 Columns: 28
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (28): sub_id, gender_female, gender_male, sub_age_MC, avg_fdmean_MC, avg...
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
neural_df <- read_csv("/Users/tur61139/Documents/GitHub/rf1-norms-hpc/n138_filterfunc.csv")
```

```
## Rows: 138 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (13): sub_id, AI_act_C11, AI_act_C12, dACC_act_C11, dACC_act_C12, mPFC_a...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df_extracted <- read_csv("/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/compiled
```

```
## Rows: 138 Columns: 45
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (45): sub_id, act_AlphaDiff-ExtBet-Int_C12_zstat3_cluster1, act_AlphaDif...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
full_df <- left_join(full_df, df, by = "sub_id")
full_df <- full_df |>
  rename(avg_ext_bet = avg_ext_bet.x,
         avg_ext_wor = avg_ext_wor.x,
         avg_int_bet = avg_int_bet.x,
         avg_int_wor = avg_int_wor.x,
         reldep = reldep.x,
         gender_female = gender_female.x,
         gender_male = gender_male.x)

# Trial-level behavioral accept/reject variables
full_df <- full_df |>
  mutate(offer = R_Option + L_Option,
         beh_fairness = offer / Endowment,
         accept = ifelse(accept_reject == "Accept", 1, 0),

         # Percentage label for plotting
         beh_fairness_perc = case_when(beh_fairness >= 0.060 & beh_fairness <= 0.0625 ~ "5%",
                                       beh_fairness >= 0.09375 & beh_fairness <= 0.125 ~ "10%",
                                       abs(beh_fairness - 0.25) < 1e-6 ~ "25%",
                                       abs(beh_fairness - 0.50) < 1e-6 ~ "50%",
                                       TRUE ~ NA_character_),
         # Numeric binned version for modeling (since )
         beh_fairness_num = case_when(beh_fairness_perc == "5%" ~ 0.05,
                                      beh_fairness_perc == "10%" ~ 0.10,
                                      beh_fairness_perc == "25%" ~ 0.25,
                                      beh_fairness_perc == "50%" ~ 0.50,
                                      TRUE ~ NA_real_),
         # Factor version for plotting
         beh_fairness_perc = factor(beh_fairness_perc, levels = c("5%", "10%", "25%", "50%")))
```

Sensitivity to fairness calculation

Calculating the 4 different slopes for sensitivities to fairness based on the behavioral data: all-trial, social trials only, nonsocial trials only, then the difference scored social - nonsocial.

```
## Sensitivity to fairness
# Function to extract subject-level fairness slope
get_fairness_slope <- function(df, label) {
  # Fixed effect of behavioral fairness, random effect with subject
  model <- glmer(accept ~ beh_fairness + (1 + beh_fairness | sub_id), data = df, family = binomial,
    control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
  fe <- fixef(model)[["beh_fairness"]]
  re <- ranef(model)$sub_id[, "beh_fairness"]
  subjIDs <- as.numeric(rownames(ranef(model)$sub_id))
  slopes <- fe + re

  tibble(sub_id = subjIDs, !!label := slopes)
}

# Estimate fairness slopes across all trials, social trials, and nonsocial trials
slope_all_df <- get_fairness_slope(full_df, "slope_all")
slope_social_df <- get_fairness_slope(filter(full_df, condition == "social"), "slope_social")
slope_nonsocial_df <- get_fairness_slope(filter(full_df, condition == "nonsocial"), "slope_nonsocial")

# Merge slopes into one df and compute social - nonsocial difference score
slope_df_wide <- reduce(list(slope_all_df, slope_social_df, slope_nonsocial_df), full_join, by = "sub_i
  mutate(diff_slope = slope_social - slope_nonsocial)
# Printing first 15 subs
print(head(slope_df_wide, 15), digits=3)
```

```
## # A tibble: 15 x 5
##    sub_id slope_all slope_social slope_nonsocial diff_slope
##     <dbl>     <dbl>        <dbl>           <dbl>      <dbl>
##  1  10369      16.1         16.6            14.0       2.61
##  2  10402      11.9         15.7            10.6       5.04
##  3  10478      35.5         30.2            40.8     -10.5
##  4  10486      10.8          6.01           13.9      -7.89
##  5  10559      22.3         35.8            15.7      20.1
##  6  10581      11.3         15.9            10.4       5.55
##  7  10584      33.7         36.3            30.2       6.13
##  8  10589      22.1         29.2            17.6      11.6
##  9  10596      22.1         31.7            19.4      12.4
## 10  10606      31.0         31.5            32.0      -0.481
## 11  10608      16.2         13.1            30.3     -17.2
## 12  10636      32.9         35.2            29.4       5.78
## 13  10640      41.6         40.8            38.2       2.62
## 14  10641      30.2         29.4            28.7       0.728
## 15  10642      15.1         13.5            18.7      -5.21
```

```
# Add slope variables back into the trial-level dataframe
full_df <- left_join(full_df, slope_df_wide, by = "sub_id")
```

Dataframe for analyses...

```r
# Change the names so that it's more legible for figures
pretty_names <- c(gender_female = "Female", gender_male = "Male", sub_age = "Age", sub_gender = "Gender
                  usi_total = "USI Score", ADI_NATRANK = "ADI (National Rank)", reldep = "Relative Depr
                  avg_ext_bet.x = "Extrinsic Improving",
                  avg_ext_wor.x = "Extrinsic Worsening",
                  avg_int_bet.x = "Intrinsic Improving",
                  avg_int_wor.x = "Intrinsic Worsening",
                  slope_all = "Sensitivity to Fairness",
                  diff_slope = "Sensitivity to Fairness, S-NS",
                  slope_social = "Social Sensitivity to Fairness",
                  slope_nonsocial = "Nonsocial Sensitivity to Fairness")

# Subject-level covariates pulled from full_df
subject_vars <- full_df |>
  select(sub_id, starts_with("avg_"), ends_with("_MC")) |>
  distinct()

# Add subject-level covariates to df used for downstream analyses
df <- left_join(df, subject_vars, by = "sub_id")

# Long-format slope dataframe for social vs. nonsocial plots
slope_df <- slope_df_wide |>
  left_join(full_df |>
              select(sub_id, avg_ext_bet, avg_ext_wor, avg_int_bet, avg_int_wor, reldep) |>
              distinct(), by = "sub_id") |>
  pivot_longer(cols = c(slope_social, slope_nonsocial), names_to = "condition", values_to = "slope") |>
  mutate(condition = ifelse(condition == "slope_social", "social", "nonsocial"))

# Subject-level acceptance rates within each condition
accept_rates <- full_df |>
  group_by(sub_id, condition) |>
  summarise(accept_rate = mean(accept, na.rm = TRUE), .groups = "drop") |>
  left_join(full_df |>
              select(sub_id, avg_ext_bet, avg_ext_wor, avg_int_bet, avg_int_wor, reldep) |>
              distinct(), by = "sub_id")

# Final df for modeling
df <- left_join(df_extracted, df, by = "sub_id")

# Create df with subject demographic and questionnaire information from long-format trial df
demog_df <- full_df |>
  group_by(sub_id) |>
  slice(1) |>
  ungroup() |>
  select(sub_id, gender_female ,gender_male, sub_age, race, ethnicity, sub_gender,
         usi_total, ADI_NATRANK, reldep, # USI and ADI and relative deprivation
         avg_ext_bet, avg_ext_wor, avg_int_bet, avg_int_wor, # EROS scores
         slope_all, diff_slope, slope_social, slope_nonsocial)
```

Behavioral Analyses

Acceptance Rate

Looking at the acceptance rate for offers across the 4 levels of fairness. In particular, want to see if there is a significant difference in 1) fairness on offer acceptance and 2) fairness and social context (human vs. computer partner) on offer acceptance.

```
fairness <- glmer(accept ~ beh_fairness_perc + (1 | sub_id), data = full_df, family = binomial)
summary(fairness)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_perc + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7518     7555    -3754     7508    12489
##
## Scaled residuals:
##    Min     1Q Median     3Q    Max
## -96.29  -0.27   0.06   0.25   7.71
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 7.51     2.74
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                       Estimate Std. Error z value             Pr(>|z|)
## (Intercept)           -1.0646     0.2436   -4.37             0.000012 ***
## beh_fairness_perc10%   0.7378     0.0809    9.12 < 0.0000000000000002 ***
## beh_fairness_perc25%   3.9793     0.0962   41.37 < 0.0000000000000002 ***
## beh_fairness_perc50%   6.0291     0.1348   44.73 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) b__10% b__25%
## bh_frns_10% -0.180
## bh_frns_25% -0.184  0.527
## bh_frns_50% -0.139  0.379  0.531
```

```
car::Anova(fairness, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: accept
##                     Chisq Df           Pr(>Chisq)
## (Intercept)          19.1  1             0.000012 ***
## beh_fairness_perc 2786.0  3 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fairness_condition <- glmer(accept ~ beh_fairness_perc * condition + (1 | sub_id), data = full_df, fami
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00511866 (tol = 0.002, component 1)
```

```r
summary(fairness_condition)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_perc * condition + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7521     7588    -3752     7503    12485
##
## Scaled residuals:
##    Min     1Q Median     3Q    Max
## -88.99  -0.28   0.06   0.26   7.62
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 7.52     2.74
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                                      Estimate Std. Error z value
## (Intercept)                           -1.0197     0.2504   -4.07
## beh_fairness_perc10%                   0.7206     0.1127    6.39
## beh_fairness_perc25%                   3.9975     0.1258   31.79
## beh_fairness_perc50%                   6.1680     0.1856   33.24
## conditionsocial                       -0.0911     0.1183   -0.77
## beh_fairness_perc10%:conditionsocial   0.0356     0.1589    0.22
## beh_fairness_perc25%:conditionsocial  -0.0315     0.1614   -0.20
## beh_fairness_perc50%:conditionsocial  -0.2517     0.2326   -1.08
##                                                 Pr(>|z|)
## (Intercept)                                 0.00004659644 ***
## beh_fairness_perc10%                        0.00000000016 ***
## beh_fairness_perc25%                       < 0.0000000000000002 ***
## beh_fairness_perc50%                       < 0.0000000000000002 ***
## conditionsocial                                     0.44
## beh_fairness_perc10%:conditionsocial                0.82
## beh_fairness_perc25%:conditionsocial                0.85
## beh_fairness_perc50%:conditionsocial                0.28
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh__10% bh__25% bh__50% cndtns b__10%: b__25%:
## bh_frns_10% -0.246
## bh_frns_25% -0.243  0.527
## bh_frns_50% -0.171  0.359   0.441
## conditinscl -0.233  0.514   0.458   0.311
## bh_frn_10%:  0.174 -0.697  -0.344  -0.233  -0.744
```

```
## bh_frn_25%:  0.171 -0.377  -0.644  -0.231  -0.733  0.545
## bh_frn_50%:  0.119 -0.262  -0.235  -0.686  -0.509  0.379   0.373
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00511866 (tol = 0.002, component 1)
```

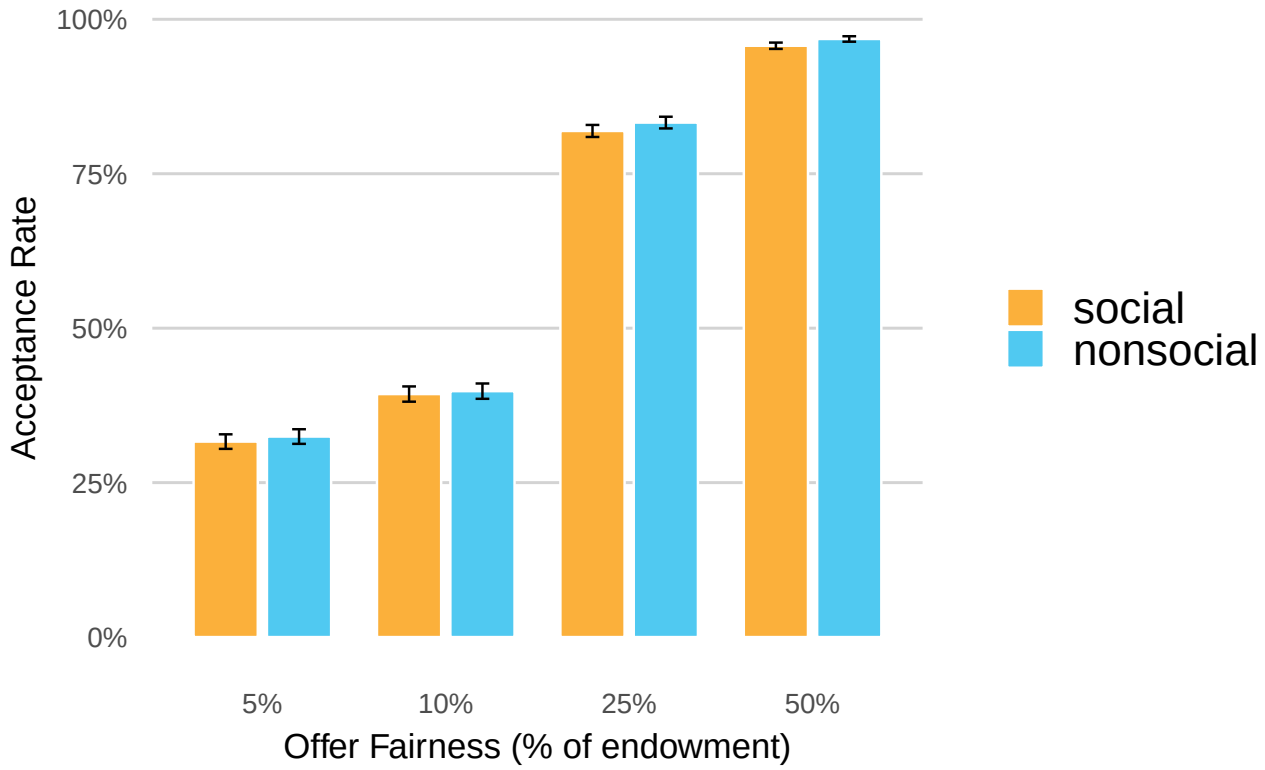
```
car::Anova(fairness_condition, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: accept
##                               Chisq Df           Pr(>Chisq)
## (Intercept)                   16.58  1             0.000047 ***
## beh_fairness_perc           1722.74  3 < 0.0000000000000002 ***
## condition                     0.59  1                 0.44
## beh_fairness_perc:condition   1.65  3                 0.65
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Summarise with SE and put "social" first
plot_df <- full_df |>
  group_by(beh_fairness_perc, condition) |>
  summarise(n = dplyr::n(),
            accept_rate = mean(accept_binary, na.rm = TRUE),
            sd_accept = sd(accept_binary, na.rm = TRUE),
            se_accept = sd_accept / sqrt(n), .groups = "drop") |>
  mutate(condition = factor(condition, levels = c("social", "nonsocial")),
         ymin = pmax(0, accept_rate - se_accept),
         ymax = pmin(1, accept_rate + se_accept))

pd <- position_dodge(width = 0.8)  # shared dodge object

ggplot(plot_df, aes(x = beh_fairness_perc, y = accept_rate, fill = condition)) +
  geom_hline(yintercept = c(0.25, 0.50, 0.75, 1.00),
             linetype = "solid",
             color = "lightgray",
             linewidth = 0.6) +
  geom_col(width = 0.7, color = "white", position = pd) +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.15, position = pd, linewidth = 0.5) +
  scale_y_continuous(name = "Acceptance Rate",
                     breaks = c(0, 0.25, 0.50, 0.75, 1.00),
                     labels = scales::label_percent(accuracy = 1)) +
  coord_cartesian(ylim = c(0, 1)) +
  scale_x_discrete(name = "Offer Fairness (% of endowment)") +
  scale_fill_manual(values = c("social" = "#FBB03B", "nonsocial" = "#50C9F1"), breaks = c("social", "no
  theme_minimal(base_size = 24) +
  theme(panel.grid = element_blank(), axis.title = element_text(size = 15), axis.text = element_text(si
```



Significant for offer fairness, but not the interaction term.

Anterior Insula Categorical Plot

Using data from the categorical model in FSL (level 1 = 5% offer, and so on) to demonstrate the anterior insula tracking offer fairness regardless of a low or high endowment amount.

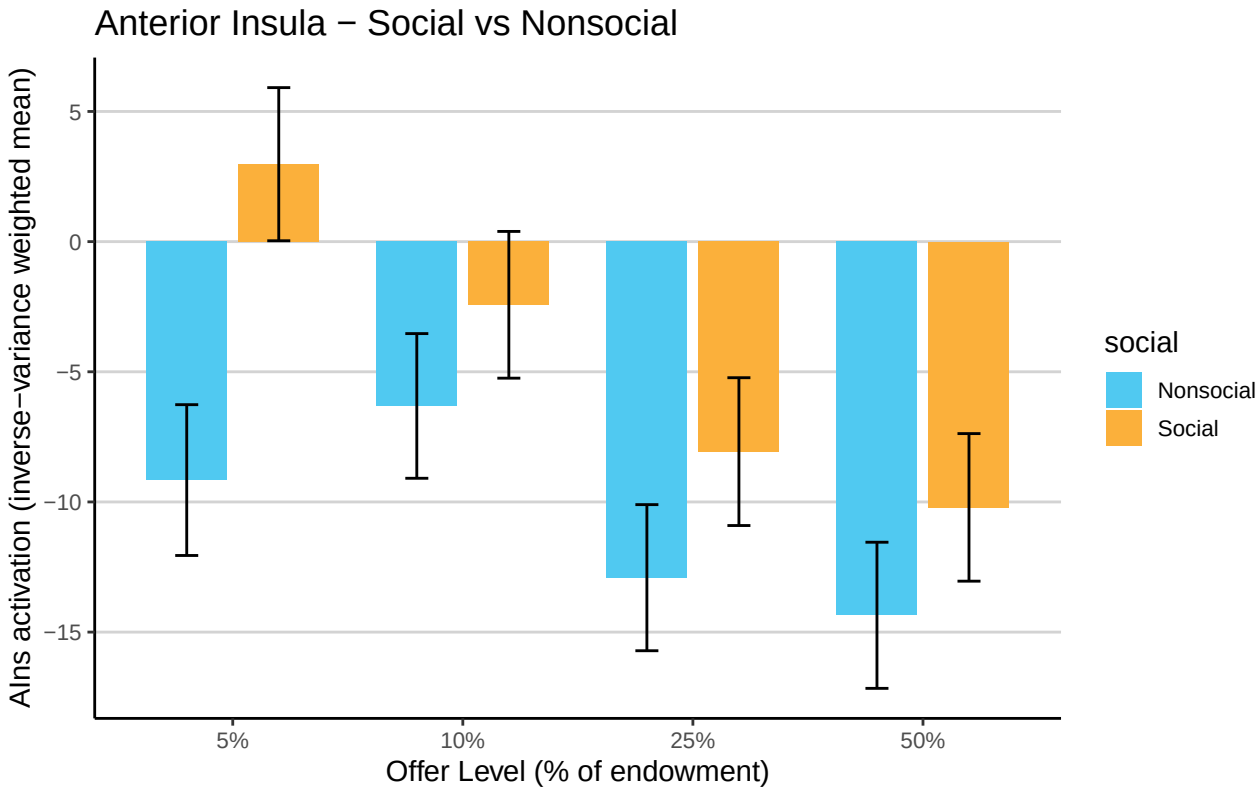
```
# 8 conditions
conditions <- c("/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty

# Weighted means and SEM function thx chat
summarise_condition <- function(cond) {
  val_file <- paste0(cond, ".txt")
  var_file <- file.path(dirname(cond), paste0("var_", basename(cond), ".txt"))
  vals <- scan(val_file, what = numeric(), quiet = TRUE)
  vars <- scan(var_file, what = numeric(), quiet = TRUE)
  if (length(vals) != length(vars)) stop("Lengths differ for ", cond)
  w <- 1 / vars
  tibble(
    condition = basename(cond),
    mean = sum(w * vals) / sum(w),
    sem  = sqrt(1 / sum(w)),
    n    = length(vals))
}
```

```r
# Weighted means and SEMs
summary_df <- map_dfr(conditions, summarise_condition)

# Parse social and level
cat_plot_df <- summary_df |>
  mutate(social = if_else(str_detect(condition, "nonsocial"), "Nonsocial", "Social"),
    level_digit = str_match(condition, "level(\\d)")[,2],
    level = factor(level_digit,
                   levels = c("1","2","3","4"),
                   labels = c("5%","10%","25%","50%")),
    cope_num = as.integer(str_match(condition, "cope-(\\d+)")[,2]))

# Plot
ggplot(cat_plot_df, aes(x = level, y = mean, fill = social)) +
  geom_hline(yintercept = c(-15, -10, -5, 0, 5),
             linetype = "solid",
             color = "lightgray",
             linewidth = 0.6) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean - sem,
                    ymax = mean + sem),
                position = position_dodge(width = 0.8), width = 0.2, linewidth=0.6) +
  scale_fill_manual(values = c("Social" = "#FBB03B", "Nonsocial" = "#50C9F1")) +
  labs(x = "Offer Level (% of endowment)",
       y = "AIns activation (inverse-variance weighted mean)",
       title = "Anterior Insula - Social vs Nonsocial") +
  theme_classic(base_size = 13)
```



```r
# Subject level df for stats
cat_stats_df <- map_dfr(conditions, function(cond) {
  val_file <- paste0(cond, ".txt")
  vals <- scan(val_file, what = numeric(), quiet = TRUE)
  tibble(sub = factor(seq_along(vals)), condition = basename(cond), value = vals)}) |>
  mutate(social = if_else(str_detect(condition, "nonsocial"), "Nonsocial", "Social"),
         level_digit = str_match(condition, "level(\\d)")[, 2],
         level = factor(level_digit, levels = c("1", "2", "3", "4"),
                        labels = c("5%", "10%", "25%", "50%")))

# Mixed model for fairness level, sociality, and interaction
m_cat <- lmer(value ~ social * level + (1 | sub), data = cat_stats_df)
anova(m_cat)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF DenDF F value       Pr(>F)
## social        12798   12798     1   959   33.22 0.0000000111 ***
## level         17276    5759     3   959   14.95 0.0000000016 ***
## social:level   3253    1084     3   959    2.81        0.038 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m_cat)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ social * level + (1 | sub)
##    Data: cat_stats_df
##
## REML criterion at convergence: 10084
##
## Scaled residuals:
##    Min     1Q Median     3Q    Max
## -5.622 -0.536  0.013  0.539  3.840
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  sub      (Intercept) 901      30.0
##  Residual             385      19.6
## Number of obs: 1104, groups:  sub, 138
##
## Fixed effects:
##                       Estimate Std. Error      df t value    Pr(>|t|)
## (Intercept)            -10.075      3.054 247.081   -3.30      0.0011 **
## socialSocial            12.738      2.363 959.000    5.39 0.000000088 ***
## level10%                 0.986      2.363 959.000    0.42      0.6765
## level25%                -4.560      2.363 959.000   -1.93      0.0539 .
## level50%                -5.977      2.363 959.000   -2.53      0.0116 *
## socialSocial:level10%   -8.321      3.342 959.000   -2.49      0.0129 *
## socialSocial:level25%   -7.801      3.342 959.000   -2.33      0.0198 *
## socialSocial:level50%   -7.592      3.342 959.000   -2.27      0.0233 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##             (Intr) sclScl lvl10% lvl25% lvl50% sS:10% sS:25%
## socialSocil -0.387
## level10%    -0.387  0.500
## level25%    -0.387  0.500  0.500
## level50%    -0.387  0.500  0.500  0.500
## sclScl:l10%  0.274 -0.707 -0.707 -0.354 -0.354
## sclScl:l25%  0.274 -0.707 -0.354 -0.707 -0.354  0.500
## sclScl:l50%  0.274 -0.707 -0.354 -0.354 -0.707  0.500  0.500
```

Activation

dlPFC, S > NS behavioral sensitivity, Extrinsic Affect Improving (Other-Directed Affect Improving)

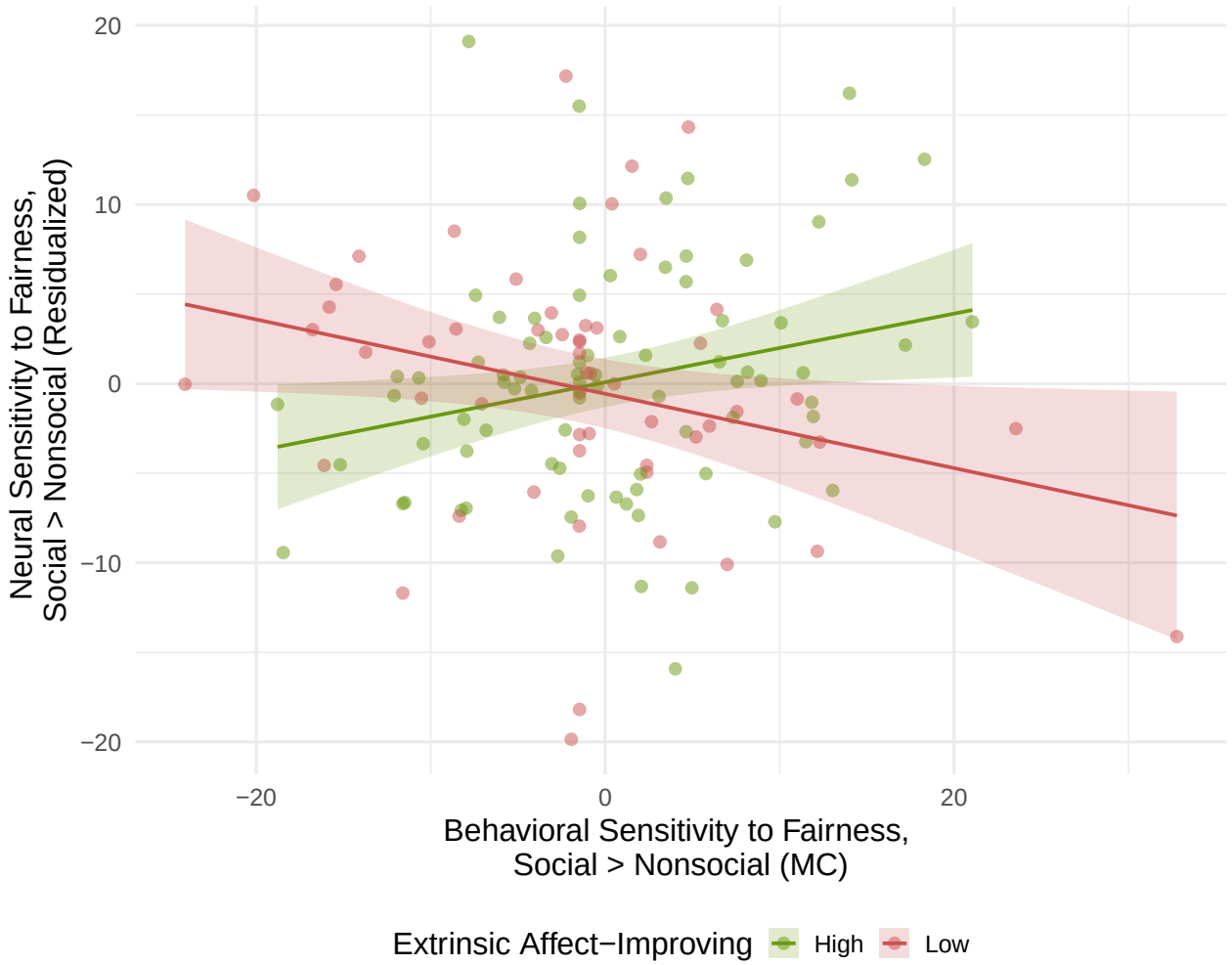
```
# Create High/Low grouping by median split
df$ext_bet_group <- ifelse(subject_vars$avg_ext_bet_MC.x >= median(subject_vars$avg_ext_bet_MC.x, na.rm
                           "High", "Low")

# Get the full 12-color color-blind friendly palette
cb_palette <- SteppedSequential5Steps

# Choose colors
cb_pair <- cb_palette[c(3, 11)]
names(cb_pair) <- c("Low", "High")


# Regression scatterplot with colorblind-safe palette and residualized X and Y
ggplot(df, aes(
  x = diff_slope,
  y = residuals(lm(`act_AlphaDiff-ExtBet-Int_C12_zstat3_cluster1` ~ sub_age_MC + gender_female + gender
  color = ext_bet_group,
  fill = ext_bet_group)) +
  geom_point(alpha = 0.5, size = 3) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.2) +
  scale_color_manual(values = cb_pair) +
  scale_fill_manual(values = cb_pair) +
  labs(
    x = "Behavioral Sensitivity to Fairness, \nSocial > Nonsocial (MC)",
    y = "Neural Sensitivity to Fairness, \nSocial > Nonsocial (Residualized)",
    color = "Extrinsic Affect-Improving",
    fill = "Extrinsic Affect-Improving") +
  theme_minimal(base_size = 18) +
  theme(legend.position="bottom")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



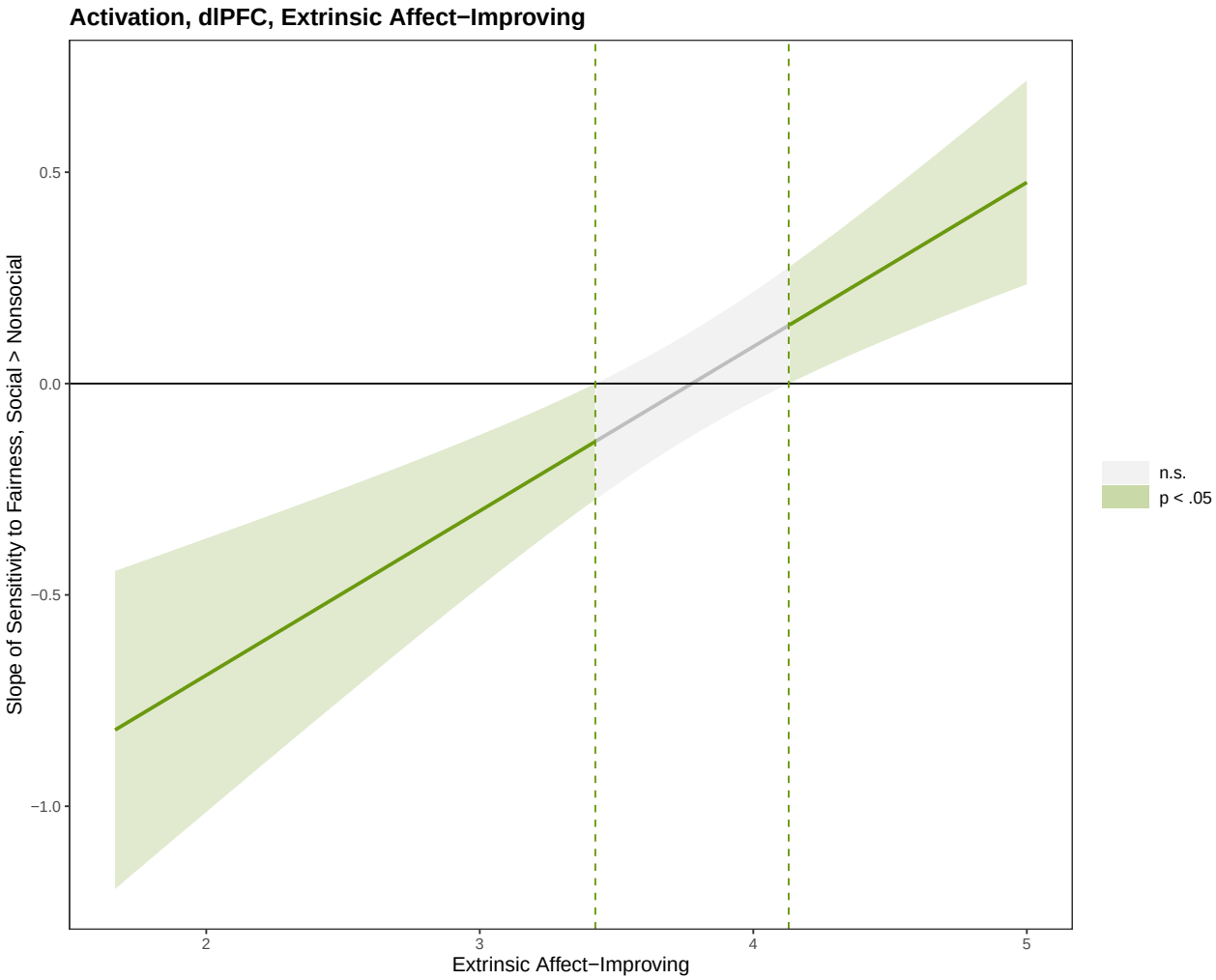

```
# Johnson-Neyman plot version
m_jn <- lm(`act_AlphaDiff-ExtBet-Int_C12_zstat3_cluster1` ~ diff_slope * avg_ext_bet.x +  sub_age_MC + g
  data = df)

# Obs range of moderator
mf <- model.frame(m_jn)
mod_rng <- range(mf$avg_ext_bet.x, na.rm = TRUE)

# Johnson-Neyman analysis
jn <- johnson_neyman(m_jn,
  pred = "diff_slope",
  modx = "avg_ext_bet.x",
  alpha = 0.05,
  mod.range = mod_rng,
  sig.color = cb_palette[11],
  insig.color = "gray",
  line.thickness = 1,
  y.label = "Slope of Sensitivity to Fairness, Social > Nonsocial",
  modx.label = "Extrinsic Affect-Improving",
  title = "Activation, dlPFC, Extrinsic Affect-Improving")
jn
```

```
## JOHNSON-NEYMAN INTERVAL
```

```
##
## When avg_ext_bet.x is OUTSIDE the interval [3.42, 4.13], the slope of
## diff_slope is p < .05.
##
## Note: The range of observed values of avg_ext_bet.x is [1.67, 5.00]
```



PPI

Amygdala, S > NS behavioral sensitivity

```r
diff_amyg <- lm(`amyg_AlphaDiff_C12` ~ sub_age_MC + gender_female + gender_male + avg_fdmean_MC + avg_ts
# NO BEHAV COVAR OF INTEREST
summary(diff_amyg)
```

```
##
## Call:
## lm(formula = amyg_AlphaDiff_C12 ~ sub_age_MC + gender_female +
##     gender_male + avg_fdmean_MC + avg_tsnr_MC, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

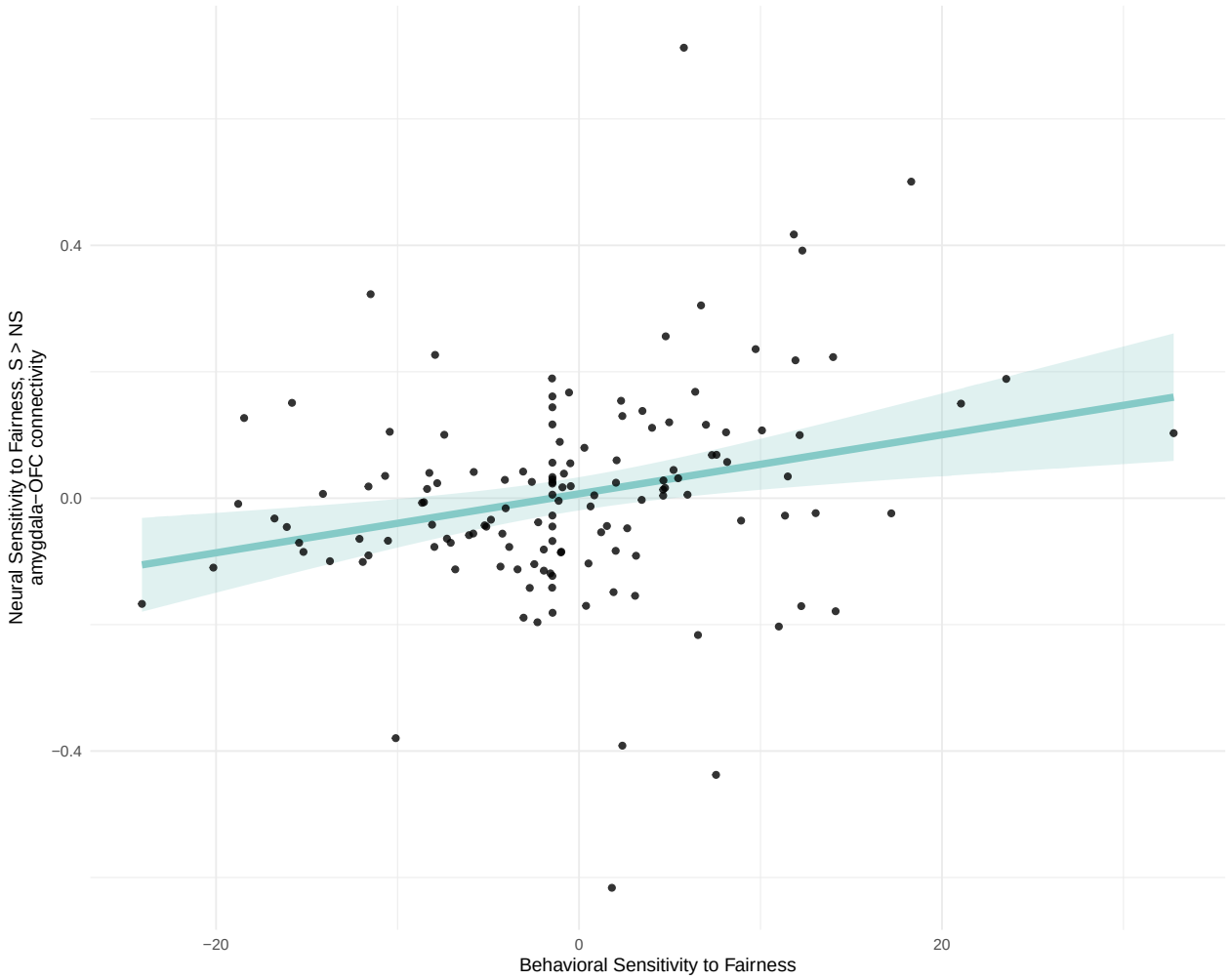
```
## -0.6162 -0.0815 -0.0045  0.0828  0.6627
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.02603    0.07355   -0.35     0.72
## sub_age_MC    -0.00164    0.00153   -1.07     0.29
## gender_female  0.02345    0.07535    0.31     0.76
## gender_male    0.04910    0.07836    0.63     0.53
## avg_fdmean_MC  0.15542    0.22784    0.68     0.50
## avg_tsnr_MC   -0.00183    0.00231   -0.79     0.43
##
## Residual standard error: 0.16 on 132 degrees of freedom
## Multiple R-squared:  0.027,  Adjusted R-squared:  -0.00982
## F-statistic: 0.733 on 5 and 132 DF,  p-value: 0.6
```

```
diff_beh <- lm(diff_slope ~ sub_age_MC + gender_female + gender_male + avg_fdmean_MC + avg_tsnr_MC, data
# BEHAV COVAR OF INTEREST AS DV
summary(diff_beh)
```

```
##
## Call:
## lm(formula = diff_slope ~ sub_age_MC + gender_female + gender_male +
##     avg_fdmean_MC + avg_tsnr_MC, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -23.317  -4.806  -0.383   5.026  31.618
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.5386     4.0917   -0.38     0.71
## sub_age_MC      0.0712     0.0853    0.83     0.41
## gender_female   0.8020     4.1917    0.19     0.85
## gender_male     1.8358     4.3591    0.42     0.67
## avg_fdmean_MC  -0.2555    12.6753   -0.02     0.98
## avg_tsnr_MC    -0.1600     0.1288   -1.24     0.22
##
## Residual standard error: 9 on 132 degrees of freedom
## Multiple R-squared:  0.0336, Adjusted R-squared:  -0.00303
## F-statistic: 0.917 on 5 and 132 DF,  p-value: 0.472
```

```
ggplot(df, aes(x = diff_slope, y = `amyg_AlphaDiff_C12`)) +
  geom_smooth(method = "lm", formula = y ~ x,
    se = TRUE, color = "#85CAC7", fill = "#85CAC7", size = 2, alpha = 0.25) +
    geom_point(color = "black", size = 1.5, alpha = 0.8) +
  labs(x = "Behavioral Sensitivity to Fairness",
       y = "Neural Sensitivity to Fairness, S > NS\namygdala-OFC connectivity") +
  theme_minimal() +
  theme(legend.position="top")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Amygdala-dmPFC, S > NS behavioral sensitivity, Extrinsic Worsening (Other-Directed Affect Worsening)

```
# amygdala
model_amyg <- lm(`amyg_AlphaDiffExtWor_C12` ~ sub_age_MC + gender_female + gender_male + avg_fdmean_MC +
summary(model_amyg)
```

```
##
## Call:
## lm(formula = amyg_AlphaDiffExtWor_C12 ~ sub_age_MC + gender_female +
##     gender_male + avg_fdmean_MC + avg_tsnr_MC + diff_slope +
##     avg_ext_wor.x + (diff_slope * avg_ext_wor.x), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23957 -0.05384  0.00557  0.05474  0.28506
##
## Coefficients:
##                           Estimate Std. Error t value  Pr(>|t|)
## (Intercept)              -0.048307   0.051753   -0.93      0.35
## sub_age_MC                0.000497   0.000917    0.54      0.59
```

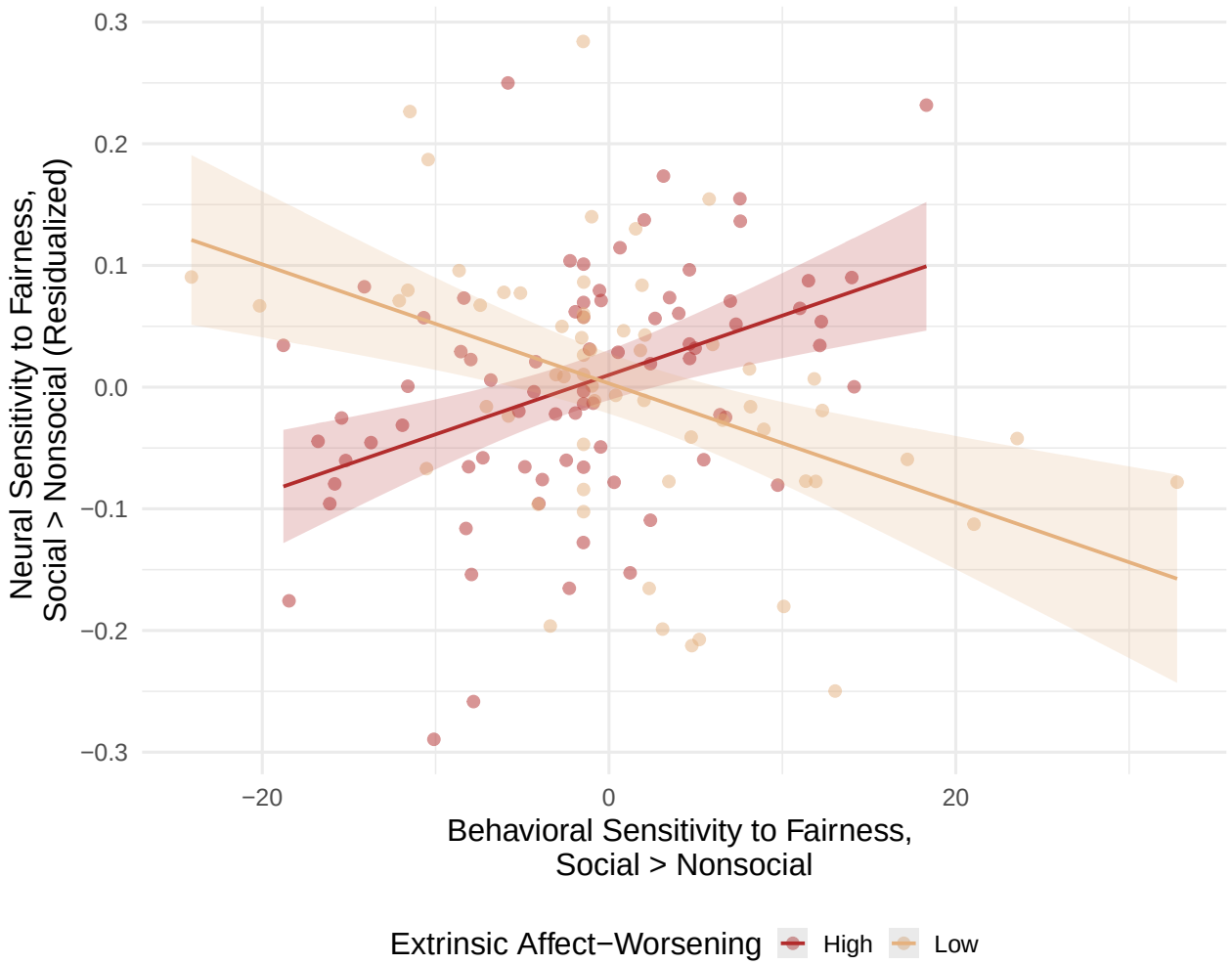
```
## gender_female             0.008836   0.044843    0.20      0.84
## gender_male               0.035407   0.046641    0.76      0.45
## avg_fdmean_MC             0.059723   0.135075    0.44      0.66
## avg_tsnr_MC               0.001796   0.001381    1.30      0.20
## diff_slope               -0.014396   0.003064   -4.70 0.0000066 ***
## avg_ext_wor.x             0.018358   0.017167    1.07      0.29
## diff_slope:avg_ext_wor.x  0.010813   0.002217    4.88 0.0000031 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.096 on 129 degrees of freedom
## Multiple R-squared:  0.167,  Adjusted R-squared:  0.115
## F-statistic: 3.23 on 8 and 129 DF,  p-value: 0.00223
```

```r
# Create High/Low grouping by median split
df$extwor_group <- ifelse(df$avg_ext_wor.x >= median(df$avg_ext_wor.x, na.rm = TRUE), "High", "Low")

cbextwor_pair <- SteppedSequential5Steps[c(2, 9)]
names(cbextwor_pair) <- c("High", "Low")

# Regression scatterplot with colorblind-safe palette and residualized Y axis
ggplot(df, aes(x = diff_slope,
               y = residuals(lm(`amyg_AlphaDiffExtWor_C12` ~ sub_age_MC + gender_female + gender_male +
               color = extwor_group,
               fill = extwor_group)) +
  geom_point(alpha = 0.5, size = 3) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.2) +
  scale_color_manual(values = cbextwor_pair) +
  scale_fill_manual(values = cbextwor_pair) +
  labs(x = "Behavioral Sensitivity to Fairness, \nSocial > Nonsocial",
       y = "Neural Sensitivity to Fairness, \nSocial > Nonsocial (Residualized)",
       color = "Extrinsic Affect-Worsening") +
  theme_minimal(base_size = 18) +
  guides(fill = "none", shape = "none") +
  theme(legend.position="bottom")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



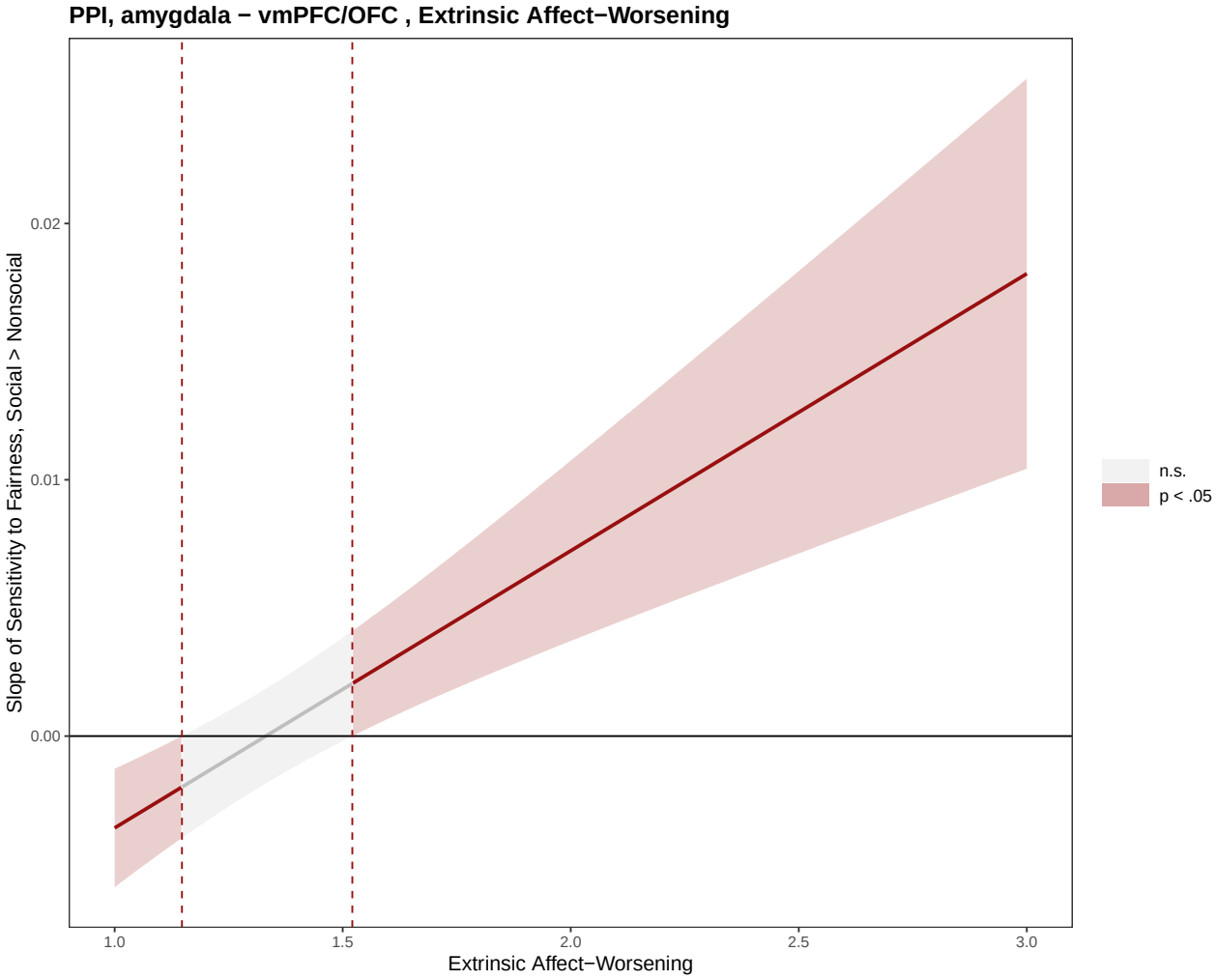

```
# Johnson-Neyman plot version
m_jn <- lm(`amyg_AlphaDiffExtWor_C12` ~ diff_slope * avg_ext_wor.x +  sub_age_MC + gender_female + gende

# Obs range of moderator
mf <- model.frame(m_jn)
mod_rng <- range(mf$avg_ext_wor.x, na.rm = TRUE)

# Johnson-Neyman analysis
jn <- johnson_neyman(m_jn,
  pred = "diff_slope",
  modx = "avg_ext_wor.x",
  alpha = 0.05,
  mod.range = mod_rng,
  sig.color = cb_palette[1],
  insig.color = "gray",
  line.thickness = 1,
  y.label = "Slope of Sensitivity to Fairness, Social > Nonsocial",
  modx.label = "Extrinsic Affect-Worsening",
  title = "PPI, amygdala - vmPFC/OFC , Extrinsic Affect-Worsening")
jn
```

```
## JOHNSON-NEYMAN INTERVAL
##
```

```
## When avg_ext_wor.x is OUTSIDE the interval [1.15, 1.52], the slope of
## diff_slope is p < .05.
##
## Note: The range of observed values of avg_ext_wor.x is [1.00, 3.00]
```



Behavioral Plots (supplement)

Overall fairness (based on task)

```
# Model with condition
glmer(accept ~ beh_fairness_num * condition + (1 | sub_id), family = binomial, full_df)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * condition + (1 | sub_id)
##    Data: full_df
##       AIC      BIC   logLik deviance df.resid
##     7783     7820    -3887     7773    12489
## Random effects:
##  Groups Name        Std.Dev.
```

```
##  sub_id (Intercept) 2.62
## Number of obs: 12494, groups:  sub_id, 138
## Fixed Effects:
##                       (Intercept)                  beh_fairness_num
##                           -1.7229                           16.3353
##                  conditionsocial  beh_fairness_num:conditionsocial
##                            0.0467                           -0.9988
```

```
summary(glmer(accept ~ beh_fairness_num * condition + (1 | sub_id), family = binomial, full_df))
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * condition + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7783     7820    -3887     7773    12489
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -147.37   -0.33    0.05    0.22    7.26
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 6.86     2.62
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                                 Estimate Std. Error z value
## (Intercept)                      -1.7229     0.2387   -7.22
## beh_fairness_num                 16.3353     0.4585   35.63
## conditionsocial                   0.0467     0.1044    0.45
## beh_fairness_num:conditionsocial -0.9988     0.5730   -1.74
##                                            Pr(>|z|)
## (Intercept)                      0.00000000000053 ***
## beh_fairness_num                < 0.0000000000000002 ***
## conditionsocial                                0.655
## beh_fairness_num:conditionsocial               0.081 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh_fr_ cndtns
## bh_frnss_nm -0.272
## conditinscl -0.225  0.545
## bh_frnss_n:  0.192 -0.680 -0.824
```

```
# Model without condition
glmer(accept ~ beh_fairness_num + (1 | sub_id), family = binomial, full_df)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num + (1 | sub_id)
```

```
##    Data: full_df
##      AIC      BIC   logLik deviance df.resid
##     7785     7808    -3890     7779    12491
## Random effects:
##  Groups Name        Std.Dev.
##  sub_id (Intercept) 2.62
## Number of obs: 12494, groups:  sub_id, 138
## Fixed Effects:
##      (Intercept)  beh_fairness_num
##             -1.7              15.8
```

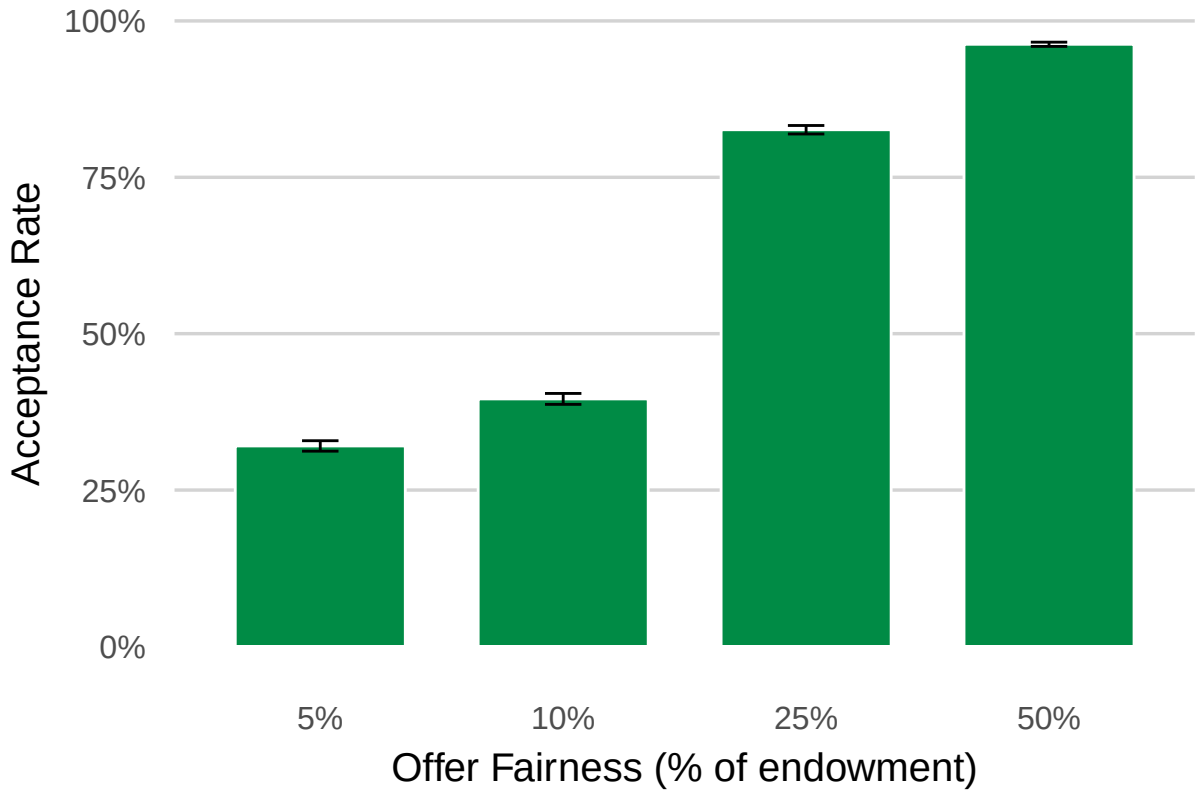
```
summary(glmer(accept ~ beh_fairness_num  + (1 | sub_id), family = binomial, full_df))
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7785     7808    -3890     7779    12491
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -163.83   -0.33    0.05    0.21    7.34
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 6.85     2.62
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                  Estimate Std. Error z value             Pr(>|z|)
## (Intercept)        -1.696      0.232    -7.3     0.00000000000029 ***
## beh_fairness_num   15.803      0.336    47.0 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## bh_frnss_nm -0.203
```

```
# Supp. Figure 1: offer fairness (task-defined, 5, 10, 25, 50% of offer)
plot_df_fairness <- full_df |>
  group_by(beh_fairness_perc) |>
  summarise(n = dplyr::n(),
    accept_rate = mean(accept_binary, na.rm = TRUE),
    sd_accept = sd(accept_binary, na.rm = TRUE),
    se_accept = sd_accept / sqrt(n),
    .groups = "drop") |>
  mutate(ymin = pmax(0, accept_rate - se_accept),
    ymax = pmin(1, accept_rate + se_accept))

ggplot(plot_df_fairness, aes(x = beh_fairness_perc, y = accept_rate)) +
  geom_hline(yintercept = c(0.25, 0.50, 0.75, 1.00),
```

```
            linetype = "solid",
            color = "lightgray",
            linewidth = 0.6) +
  geom_col(width = 0.7, fill = "springgreen4", color = "white") +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.15, linewidth = 0.5) +
  scale_y_continuous(name = "Acceptance Rate",
                     breaks = c(0, 0.25, 0.50, 0.75, 1.00),
                     labels = scales::label_percent(accuracy = 1)) +
  coord_cartesian(ylim = c(0, 1)) +
  scale_x_discrete(name = "Offer Fairness (% of endowment)") +
  theme_minimal(base_size = 24) +
  theme(panel.grid = element_blank(), axis.title = element_text(size = 15),
        axis.text = element_text(size = 12),
        legend.position = "none")
```




```
# EROS moderation of fairness x offer acceptance relationship
fair_extwor <- glmer(accept ~ beh_fairness_num * avg_ext_wor + (1 | sub_id), family = binomial, data = f
fair_extwor
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_ext_wor + (1 | sub_id)
##    Data: full_df
##       AIC      BIC   logLik deviance df.resid
##     7716     7753    -3853     7706    12489
## Random effects:
##  Groups Name        Std.Dev.
##  sub_id (Intercept) 2.59
```

```
## Number of obs: 12494, groups:  sub_id, 138
## Fixed Effects:
##                   (Intercept)              beh_fairness_num
##                         0.845                         7.258
##                   avg_ext_wor  beh_fairness_num:avg_ext_wor
##                        -1.860                         6.291
```

```
summary(fair_extwor)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_ext_wor + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7716     7753    -3853     7706    12489
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -188.15   -0.32    0.04    0.22   16.85
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 6.69     2.59
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                              Estimate Std. Error z value           Pr(>|z|)
## (Intercept)                     0.845      0.701    1.20               0.23
## beh_fairness_num                7.258      1.080    6.72 0.0000000000183524 ***
## avg_ext_wor                    -1.860      0.476   -3.91 0.0000941808171833 ***
## beh_fairness_num:avg_ext_wor    6.291      0.802    7.85 0.0000000000000043 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh_fr_ avg_x_
## bh_frnss_nm -0.243
## avg_ext_wor -0.945  0.257
## bh_frns_:__  0.237 -0.952 -0.273
```

```
fair_extbet <- glmer(accept ~ beh_fairness_num * avg_ext_bet + (1 | sub_id), family = binomial, data = 
fair_extbet
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_ext_bet + (1 | sub_id)
##    Data: full_df
##      AIC      BIC   logLik deviance df.resid
##     7766     7803    -3878     7756    12489
## Random effects:
##  Groups Name        Std.Dev.
##  sub_id (Intercept) 2.64
```

```
## Number of obs: 12494, groups:  sub_id, 138
## Fixed Effects:
##                   (Intercept)              beh_fairness_num
##                        -3.583                        24.911
##                   avg_ext_bet  beh_fairness_num:avg_ext_bet
##                         0.468                        -2.257
```

```
summary(fair_extbet)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_ext_bet + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7766     7803    -3878     7756    12489
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -180.66   -0.33    0.05    0.21    7.96
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 6.95     2.64
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                              Estimate Std. Error z value             Pr(>|z|)
## (Intercept)                    -3.583      1.253   -2.86               0.0042
## beh_fairness_num               24.911      1.956   12.73 < 0.0000000000000002
## avg_ext_bet                     0.468      0.309    1.51               0.1301
## beh_fairness_num:avg_ext_bet   -2.257      0.469   -4.81            0.0000015
##
## (Intercept)                  **
## beh_fairness_num             ***
## avg_ext_bet
## beh_fairness_num:avg_ext_bet ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh_fr_ avg_x_
## bh_frnss_nm -0.230
## avg_ext_bet -0.982  0.220
## bh_frns_:__  0.227 -0.985 -0.223
```

```
fair_intwor <- glmer(accept ~ beh_fairness_num * avg_int_wor + (1 | sub_id), family = binomial, data = f
fair_intwor
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_int_wor + (1 | sub_id)
##    Data: full_df
```

```
##      AIC      BIC   logLik deviance df.resid
##     7767     7804    -3878     7757    12489
## Random effects:
##  Groups Name        Std.Dev.
##  sub_id (Intercept) 2.58
## Number of obs: 12494, groups:  sub_id, 138
## Fixed Effects:
##                  (Intercept)              beh_fairness_num
##                        0.291                        10.164
##                  avg_int_wor  beh_fairness_num:avg_int_wor
##                       -0.769                         2.182
```

```
summary(fair_intwor)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_int_wor + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7767     7804    -3878     7757    12489
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -157.05   -0.32    0.05    0.23    7.75
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 6.64     2.58
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                              Estimate Std. Error z value            Pr(>|z|)
## (Intercept)                     0.291      0.872    0.33               0.738
## beh_fairness_num               10.164      1.245    8.17 0.00000000000000032
## avg_int_wor                    -0.769      0.324   -2.37               0.018
## beh_fairness_num:avg_int_wor    2.182      0.478    4.57 0.00000492975063481
##
## (Intercept)
## beh_fairness_num             ***
## avg_int_wor                  *
## beh_fairness_num:avg_int_wor ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh_fr_ avg_n_
## bh_frnss_nm -0.196
## avg_int_wor -0.965  0.195
## bh_frns_:__  0.189 -0.963 -0.203
```

```
fair_intbet <- glmer(accept ~ beh_fairness_num * avg_int_bet + (1 | sub_id), family = binomial, data = f
fair_intbet
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_int_bet + (1 | sub_id)
##    Data: full_df
##      AIC      BIC   logLik deviance df.resid
##     7788     7825    -3889     7778    12489
## Random effects:
##  Groups Name        Std.Dev.
##  sub_id (Intercept) 2.62
## Number of obs: 12494, groups:  sub_id, 138
## Fixed Effects:
##                  (Intercept)              beh_fairness_num
##                       -1.091                        17.013
##                  avg_int_bet  beh_fairness_num:avg_int_bet
##                       -0.209                        -0.412
```

```
summary(fair_intbet)
```

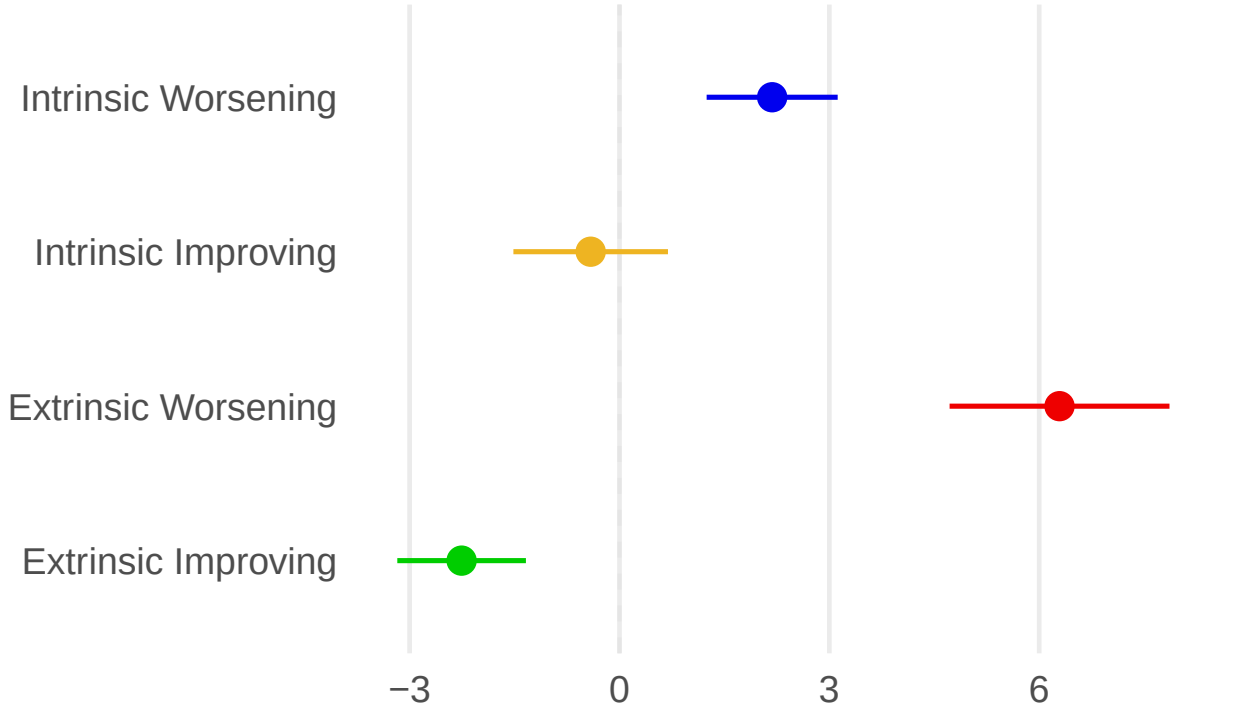
```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_num * avg_int_bet + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7788     7825    -3889     7778    12489
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -173.79   -0.33    0.05    0.21    7.52
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 6.84     2.62
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                              Estimate Std. Error z value            Pr(>|z|)
## (Intercept)                    -1.091      1.111   -0.98                0.33
## beh_fairness_num               17.013      1.697   10.03 <0.0000000000000002
## avg_int_bet                    -0.209      0.374   -0.56                0.58
## beh_fairness_num:avg_int_bet   -0.412      0.564   -0.73                0.47
##
## (Intercept)
## beh_fairness_num             ***
## avg_int_bet
## beh_fairness_num:avg_int_bet
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh_fr_ avg_n_
## bh_frnss_nm -0.206
## avg_int_bet -0.978  0.200
```

```
## bh_frns_:__  0.202 -0.980 -0.204
```

```
# Supp Figure 2: EROS moderation coefficient plot...
eros_models <- list(
  "Extrinsic Improving" = fair_extbet,
  "Extrinsic Worsening" = fair_extwor,
  "Intrinsic Improving" = fair_intbet,
  "Intrinsic Worsening" = fair_intwor
)
fig2_df <- purrr::imap_dfr(eros_models, \(model, scale_name) {
  broom.mixed::tidy(model, effects = "fixed", conf.int = TRUE) |>
    dplyr::filter(stringr::str_detect(term, "beh_fairness_num:")) |>
    dplyr::mutate(EROS_scale = scale_name)
}) |>
  dplyr::mutate(
    EROS_scale = factor(EROS_scale,
      levels = c("Extrinsic Improving", "Extrinsic Worsening",
        "Intrinsic Improving", "Intrinsic Worsening")))

eros_cols <- c("Extrinsic Improving" = "green3",
               "Extrinsic Worsening" = "red2",
               "Intrinsic Improving" = "goldenrod2",
               "Intrinsic Worsening" = "blue2")

ggplot(fig2_df, aes(x = EROS_scale, y = estimate, color = EROS_scale)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray90",
    linewidth = 0.7) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
    linewidth = 0.9,
    size = 1) +
  coord_flip() +
  scale_color_manual(values = eros_cols) +
  guides(color = "none") +
  labs(x = NULL,
       y = "Offer Fairness and EROS Subscale Interaction Estimates",
       caption = "Points represent GLMM interaction estimates; error bars represent 95% CIs.") +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_blank(),
    axis.text = element_text(size = 14),
    axis.title = element_text(size = 16)
  )
```



Offer Fairness and EROS Subscale Interaction Estim

)ints represent GLMM interaction estimates; error bars represent 95% CIs.

Subject-specific fairness slopes

```
ext_wor_all <- summary(lm(slope_all ~ avg_ext_wor.x, df))
ext_wor_all
```

```
##
## Call:
## lm(formula = slope_all ~ avg_ext_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -23.35  -9.35   1.27   8.35  26.08
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      11.90       2.93    4.06 0.000082 ***
## avg_ext_wor.x     4.35       1.98    2.20    0.029 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11 on 136 degrees of freedom
## Multiple R-squared:  0.0344, Adjusted R-squared:  0.0273
## F-statistic: 4.84 on 1 and 136 DF,  p-value: 0.0295
```

```
ext_bet_all <- summary(lm(slope_all ~ avg_ext_bet.x, df))
ext_bet_all
```

```
##
```

```
## Call:
## lm(formula = slope_all ~ avg_ext_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -24.18 -10.74   1.63   8.51  26.09
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      24.67       5.20    4.74 0.0000053 ***
## avg_ext_bet.x    -1.69       1.29   -1.31      0.19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0125, Adjusted R-squared:  0.0052
## F-statistic: 1.72 on 1 and 136 DF,  p-value: 0.192
```

```
int_wor_all <- summary(lm(slope_all ~ avg_int_wor.x, df))
int_wor_all
```

```
##
## Call:
## lm(formula = slope_all ~ avg_int_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -24.86 -10.13   1.51   8.64  27.58
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      11.73       3.71    3.17   0.0019 **
## avg_int_wor.x     2.41       1.38    1.75   0.0832 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11 on 136 degrees of freedom
## Multiple R-squared:  0.0219, Adjusted R-squared:  0.0147
## F-statistic: 3.05 on 1 and 136 DF,  p-value: 0.0832
```

```
int_bet_all <- summary(lm(slope_all ~ avg_int_bet.x, df))
int_bet_all
```

```
##
## Call:
## lm(formula = slope_all ~ avg_int_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -25.09 -11.11   2.53   8.71  24.31
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     18.103      4.718    3.84  0.00019 ***
## avg_int_bet.x   -0.044      1.590   -0.03  0.97797
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  5.63e-06,   Adjusted R-squared:  -0.00735
## F-statistic: 0.000765 on 1 and 136 DF,  p-value: 0.978
```

```
ext_wor_diff <- summary(lm(diff_slope ~ avg_ext_wor.x, df))
ext_wor_diff
```

```
##
## Call:
## lm(formula = diff_slope ~ avg_ext_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -24.55  -4.83  -0.77   4.92  32.31
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)       2.69       2.31    1.17     0.24
## avg_ext_wor.x    -2.23       1.56   -1.43     0.15
##
## Residual standard error: 9 on 136 degrees of freedom
## Multiple R-squared:  0.0149, Adjusted R-squared:  0.00767
## F-statistic: 2.06 on 1 and 136 DF,  p-value: 0.154
```

```
ext_bet_diff <- summary(lm(diff_slope ~ avg_ext_bet.x, df))
ext_bet_diff
```

```
##
## Call:
## lm(formula = diff_slope ~ avg_ext_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -23.63  -5.29  -0.89   5.08  33.34
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -0.827      4.080    -0.2     0.84
## avg_ext_bet.x    0.101      1.010     0.1     0.92
##
## Residual standard error: 9 on 136 degrees of freedom
## Multiple R-squared:  7.28e-05,   Adjusted R-squared:  -0.00728
## F-statistic: 0.00991 on 1 and 136 DF,  p-value: 0.921
```

```
int_wor_diff <- summary(lm(diff_slope ~ avg_int_wor.x, df))
int_wor_diff
```

```
##
## Call:
## lm(formula = diff_slope ~ avg_int_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
```

```
## -23.62  -5.52  -0.57   5.12  33.71
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -1.423      2.920   -0.49     0.63
## avg_int_wor.x    0.384      1.087    0.35     0.72
##
## Residual standard error: 9 on 136 degrees of freedom
## Multiple R-squared:  0.000915,   Adjusted R-squared:  -0.00643
## F-statistic: 0.125 on 1 and 136 DF,  p-value: 0.725
```

```
int_bet_diff <- summary(lm(diff_slope ~ avg_int_bet.x, df))
int_bet_diff
```

```
##
## Call:
## lm(formula = diff_slope ~ avg_int_bet.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -24.156  -4.836  -0.422   5.486  30.828
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.19       3.66    0.87     0.39
## avg_int_bet.x    -1.25       1.24   -1.01     0.31
##
## Residual standard error: 9 on 136 degrees of freedom
## Multiple R-squared:  0.00742,    Adjusted R-squared:  0.000125
## F-statistic: 1.02 on 1 and 136 DF,  p-value: 0.315
```

```
ext_wor_soc <- summary(lm(slope_social ~ avg_ext_wor.x, df))
ext_wor_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_ext_wor.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.433 -10.343   0.863  10.780  21.159
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      15.18       3.13    4.86 0.0000032 ***
## avg_ext_wor.x     2.86       2.11    1.36      0.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0133, Adjusted R-squared:  0.00607
## F-statistic: 1.84 on 1 and 136 DF,  p-value: 0.178
```

```
ext_bet_soc <- summary(lm(slope_social ~ avg_ext_bet.x, df))
ext_bet_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_ext_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -28.68 -11.41   1.99  10.61  21.74
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      25.80       5.50    4.69 0.0000065 ***
## avg_ext_bet.x    -1.67       1.36   -1.23      0.22
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0109, Adjusted R-squared:  0.00364
## F-statistic:  1.5 on 1 and 136 DF,  p-value: 0.223
```

```
int_wor_soc <- summary(lm(slope_social ~ avg_int_wor.x, df))
int_wor_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_int_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -29.35 -10.94   0.54   9.85  23.30
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      12.86       3.92    3.28   0.0013 **
## avg_int_wor.x     2.44       1.46    1.67   0.0965 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0202, Adjusted R-squared:  0.013
## F-statistic:  2.8 on 1 and 136 DF,  p-value: 0.0965
```

```
int_bet_soc <- summary(lm(slope_social ~ avg_int_bet.x, df))
int_bet_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_int_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -29.77 -11.75   1.56  11.12  20.67
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     21.653      4.979    4.35 0.000027 ***
```

```
## avg_int_bet.x   -0.852      1.678   -0.51     0.61
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.00189,    Adjusted R-squared:  -0.00545
## F-statistic: 0.257 on 1 and 136 DF,  p-value: 0.613
```

```
ext_wor_ns <- summary(lm(slope_nonsocial ~ avg_ext_wor.x, df))
ext_wor_ns
```

```
##
## Call:
## lm(formula = slope_nonsocial ~ avg_ext_wor.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -25.258 -10.807  -0.409  10.655  22.620
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      12.49       2.96    4.22 0.000045 ***
## avg_ext_wor.x     5.09       2.00    2.55    0.012 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0455, Adjusted R-squared:  0.0385
## F-statistic: 6.49 on 1 and 136 DF,  p-value: 0.012
```

```
ext_bet_ns <- summary(lm(slope_nonsocial ~ avg_ext_bet.x, df))
ext_bet_ns
```

```
##
## Call:
## lm(formula = slope_nonsocial ~ avg_ext_bet.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -26.344 -10.643  -0.276  10.102  22.419
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      26.63       5.29    5.03 0.0000015 ***
## avg_ext_bet.x    -1.77       1.31   -1.35      0.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0132, Adjusted R-squared:  0.00597
## F-statistic: 1.82 on 1 and 136 DF,  p-value: 0.179
```

```
int_wor_ns <- summary(lm(slope_nonsocial ~ avg_int_wor.x, df))
int_wor_ns
```

```
##
```

```
## Call:
## lm(formula = slope_nonsocial ~ avg_int_wor.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.098 -10.243  -0.585  10.467  23.352
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      14.28       3.78    3.77  0.00024 ***
## avg_int_wor.x     2.06       1.41    1.46  0.14644
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0154, Adjusted R-squared:  0.0082
## F-statistic: 2.13 on 1 and 136 DF,  p-value: 0.146
```

```
int_bet_ns <- summary(lm(slope_nonsocial ~ avg_int_bet.x, df))
int_bet_ns
```

```
##
## Call:
## lm(formula = slope_nonsocial ~ avg_int_bet.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.195 -10.644  -0.725  11.137  20.946
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     18.468      4.800    3.85  0.00018 ***
## avg_int_bet.x    0.394      1.618    0.24  0.80797
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.000436,   Adjusted R-squared:  -0.00691
## F-statistic: 0.0593 on 1 and 136 DF,  p-value: 0.808
```

```
# Difference between social and nonsocial context, extrinsic worsening trait in behavior (no difference
df_long_beh <- data.frame(slope = c(df$slope_social, df$slope_nonsocial),
  eros  = c(df$avg_ext_wor.x, df$avg_ext_wor.x),
  context = factor(rep(c("social", "nonsocial"), each = nrow(df))))

model_contextdiff <- lm(slope ~ eros * context, df_long_beh)
summary(model_contextdiff)
```

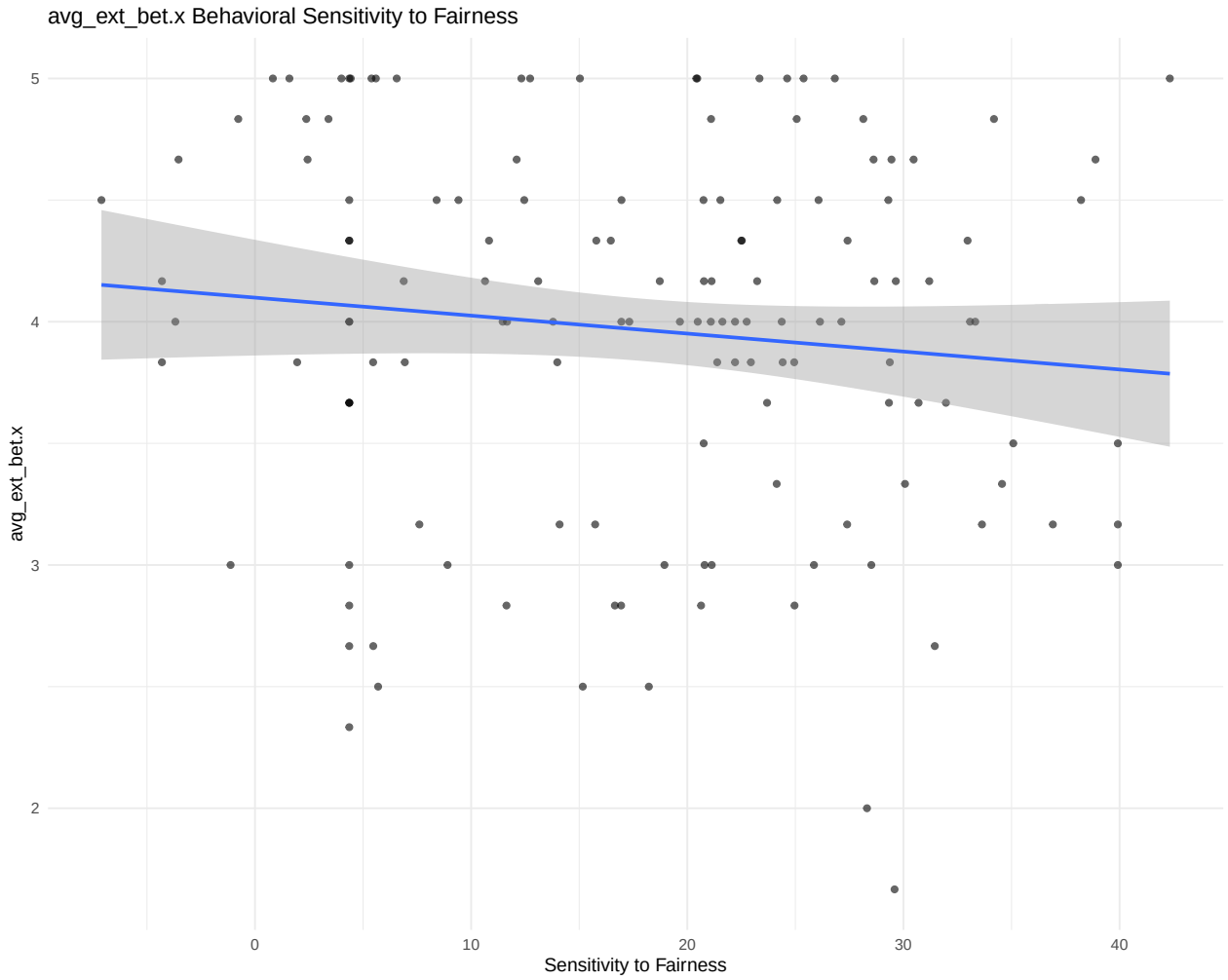
```
##
## Call:
## lm(formula = slope ~ eros * context, data = df_long_beh)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.433 -10.749   0.118  10.753  22.620
```

```
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)            12.49       3.05    4.10 0.000054 ***
## eros                    5.09       2.06    2.48    0.014 *
## contextsocial           2.69       4.31    0.63    0.532
## eros:contextsocial     -2.23       2.91   -0.77    0.443
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 272 degrees of freedom
## Multiple R-squared:  0.0291, Adjusted R-squared:  0.0184
## F-statistic: 2.72 on 3 and 272 DF,  p-value: 0.0448
```

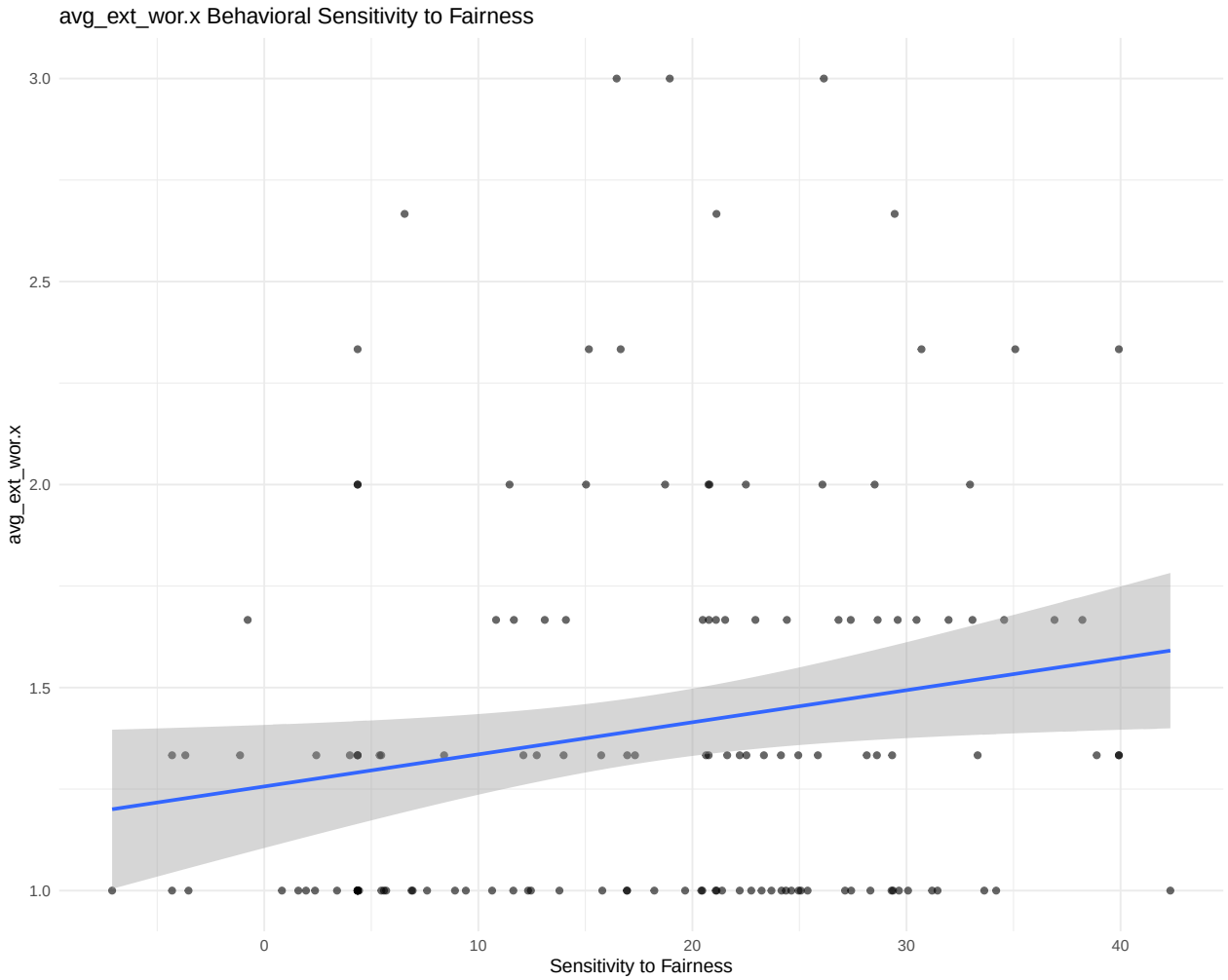
```
vars <- c("avg_ext_bet.x", "avg_ext_wor.x","avg_int_bet.x", "avg_int_wor.x")

for (v in vars) {
  p <- ggplot(df, aes(x = slope_all, y = .data[[v]])) +
    geom_point(alpha = 0.6) +
    geom_smooth(method = "lm", se = TRUE) +
    theme_minimal() +
    labs(title = paste(v, "Behavioral Sensitivity to Fairness"),
      x = "Sensitivity to Fairness",
      y = v)
    print(p)}
```

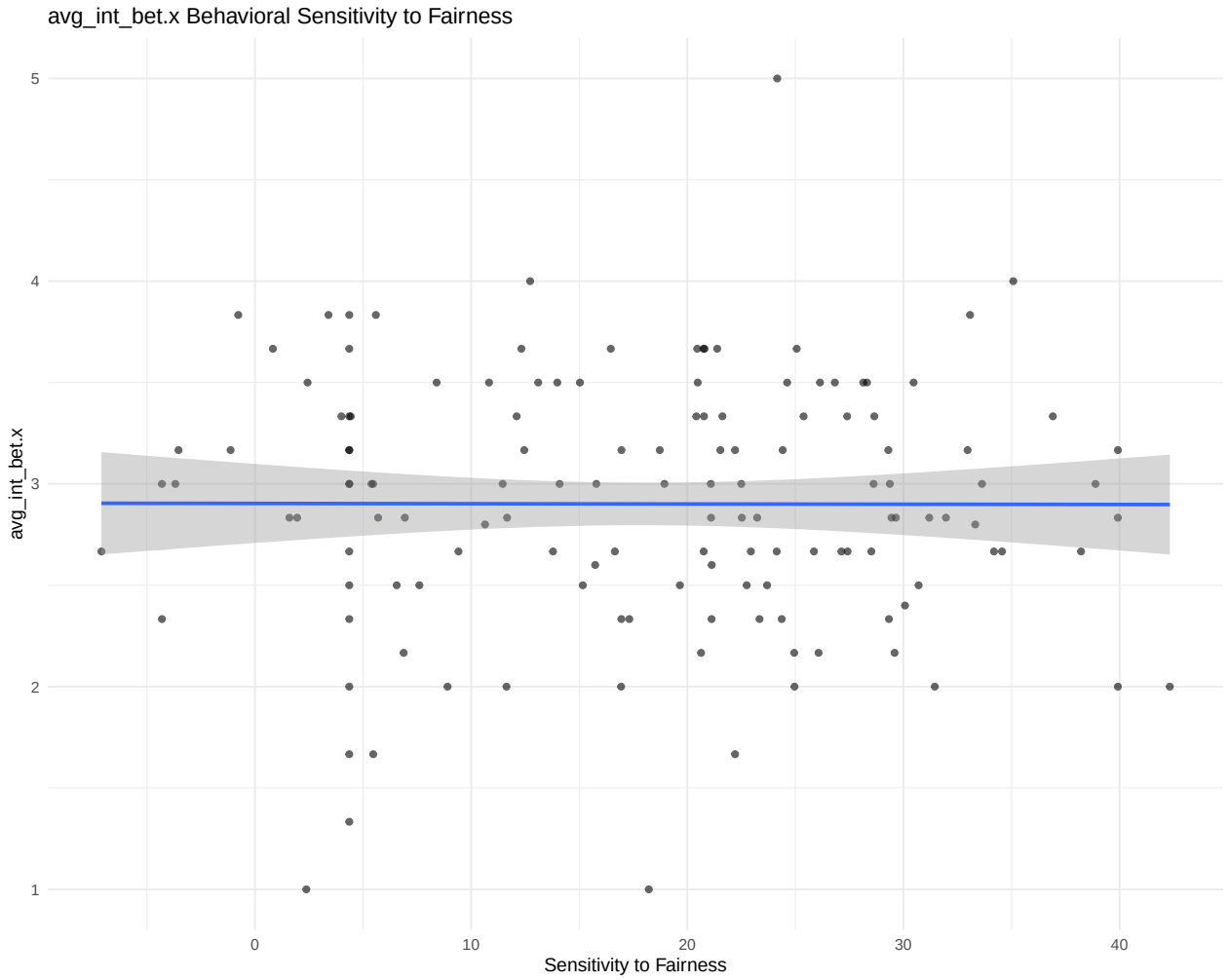
```
## `geom_smooth()` using formula = 'y ~ x'
```



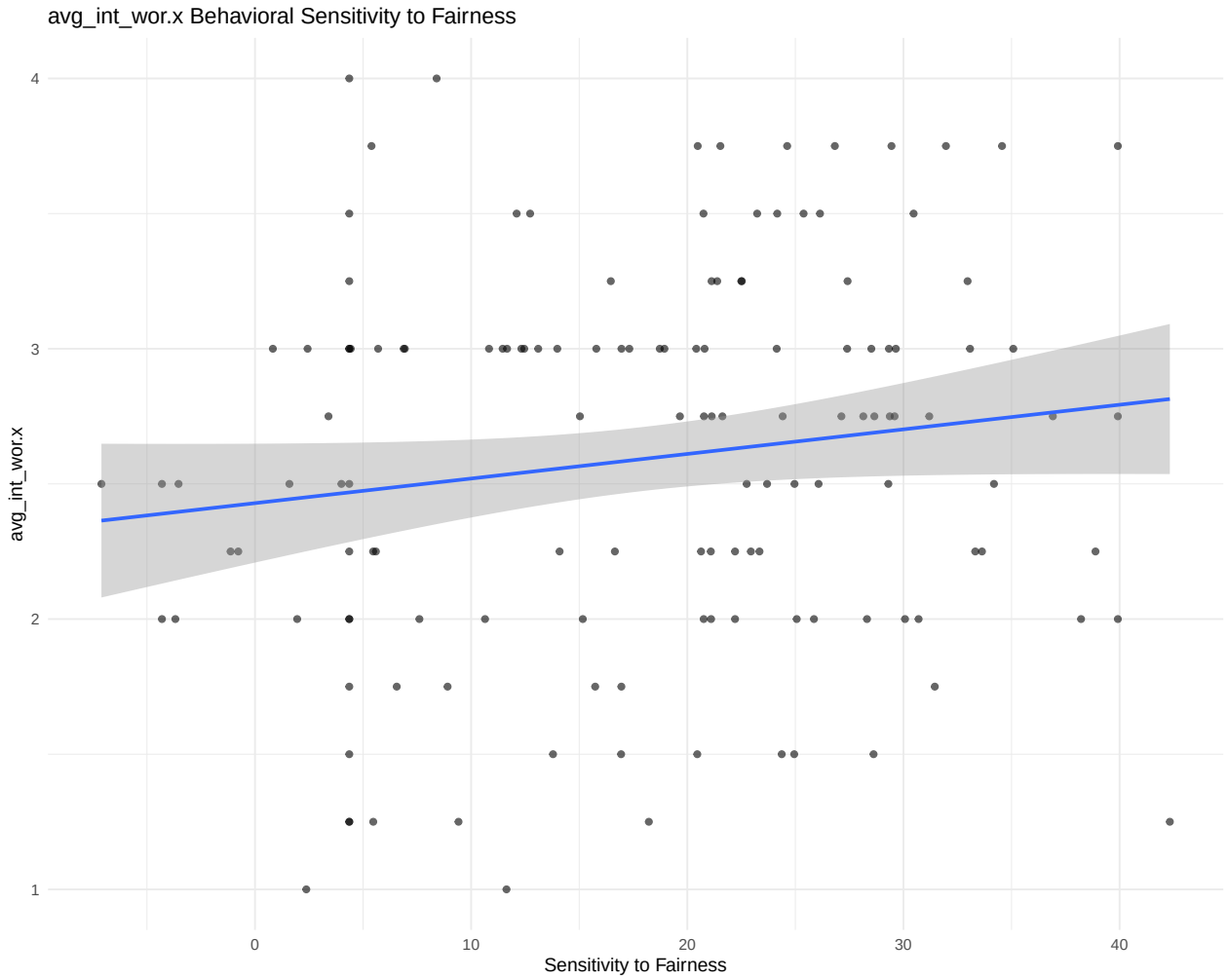

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
df_long <- df |>
  select(slope_all, avg_ext_bet.x, avg_ext_wor.x, avg_int_bet.x, avg_int_wor.x) |>
  pivot_longer(-slope_all, names_to = "EROS_scale", values_to = "score")

df_long$EROS_scale <- factor(df_long$EROS_scale,
  levels = c("avg_ext_bet.x", "avg_ext_wor.x", "avg_int_bet.x", "avg_int_wor.x"),
  labels = c("Extrinsic Improving", "Extrinsic Worsening", "Intrinsic Improving", "Intrinsic Worsening"]

# Supp. Figure 3: Subject-specific sensitivity to fairness x EROS
ggplot(df_long, aes(x = slope_all, y = score)) +
  geom_smooth(aes(color = EROS_scale, fill = EROS_scale), method = "lm", se = TRUE, linewidth = 1.2, al
  geom_point(color = "black", alpha = 0.8) +
  facet_wrap(~ EROS_scale) +
  scale_color_manual(values = c("Extrinsic Improving" = "green3",
                                "Extrinsic Worsening" = "red2",
                                "Intrinsic Improving" = "goldenrod2",
                                "Intrinsic Worsening" = "blue2")) +
  scale_fill_manual(values = c("Extrinsic Improving" = "green3",
                               "Extrinsic Worsening" = "red2",
                               "Intrinsic Improving" = "goldenrod2",
                               "Intrinsic Worsening" = "blue2")) +
  guides(color = "none", fill = "none") +
```

```
  labs(x = "Sensitivity to Fairness",
       y = "EROS Subscale Score",
       caption = "Shaded regions represent 1 standard error of the estimated regression line.") +
  theme_minimal(base_size = 16)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

